

Group: _____ **Section:** _____

Member 1: _____ **BUID:** _____

Member 2: _____ **BUID:** _____

Member 3: _____ **BUID:** _____

Member 4: _____ **BUID:** _____

Project 2 Real Data Analysis

Purpose

Today you will apply the Project 2 inference comparison to a real data set and connect real-data diagnostics to the simulation study.

Shared Group Workflow

Work in one shared Google Doc and one shared analysis folder. Assign the four roles below, then have every member review the real-data analysis and its connection to the simulations.

- **Member 1 – Data lead:** import, inspect, and document the data and variables.
- **Member 2 – OLS lead:** fit the planned model and interpret the target coefficient or prediction.
- **Member 3 – Bootstrap lead:** compute bootstrap intervals and compare them with classical intervals.
- **Member 4 – Diagnostics editor:** connect assumptions, diagnostics, and conclusions in the shared report.

In-Class Work

- (A) **Load and explore.** Import the real data set and inspect the response, predictors, missing values, and variable types.
- (B) **Fit OLS.** Fit the planned regression model and interpret the main coefficient or prediction target.
- (C) **Classical versus bootstrap.** Compare classical and bootstrap intervals for the target quantity.
- (D) **Diagnostics.** Check residual patterns and connect them to assumptions from the simulation design.
- (E) **Check-in.** Do the real-data diagnostics resemble one of your simulation scenarios?

Deliverable

Submit the Project 2 progress report through Gradescope or Blackboard by Tuesday, November 3 at 10:00 PM. Include the real-data analysis, diagnostics, shared Google Doc link, and reproducible code files.